

Exponentials and logarithms - Set 2 - Solutions

Specification MM5.1-MM5.3.

Answer key

Question	Answer	Topic
1	E	Exponentials and logarithms
2	B	Exponentials and logarithms
3	A	Exponentials and logarithms
4	B	Exponentials and logarithms
5	A	Exponentials and logarithms
6	E	Exponentials and logarithms
7	D	Exponentials and logarithms
8	E	Exponentials and logarithms
9	B	Exponentials and logarithms

Worked solutions

Question 1 LOG-16 · Exponentials and logarithms · spec MM5.1

For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

Answer: E

$x = 0$: $1 < 2$ true. $x = 1$: $3 < 4$ true. $x = 2$: $9 < 8$ false. So I and II.

Full method - how and why

Three numerical tests. Nothing conceptual - just careful arithmetic with small powers.

1 I: $x = 0$ gives $3^0 = 1$ and $2^1 = 2$. Is $1 < 2$? Yes. **True.**

why: $a^0 = 1$ for any a .

2 II: $x = 1$ gives $3^1 = 3$ and $2^2 = 4$. Is $3 < 4$? Yes. **True.**

why: Careful with the index $x + 1 = 2$.

3 III: $x = 2$ gives $3^2 = 9$ and $2^3 = 8$. Is $9 < 8$? No. **False.**

why: 3^x has overtaken 2^{x+1} by here - exponential growth with the larger base always wins eventually.

4 I and II only.

why: Match to the option list.

Where it goes wrong

- Computing 2^{x+1} as $2^x + 1$.
- Assuming the inequality holds for all x because it holds for the first two.

Takeaway Substitute each value carefully; a larger base always overtakes eventually, whatever the head start.

Question 2 LOG-18 · Exponentials and logarithms · spec MM5.2

If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

Answer: B

$b = a^2$ and $c = b^3 = a^6$, so $\log_a c = 6$.

Full method - how and why

A chain of logarithms. Convert each statement to exponential form and substitute.

1 $\log_a b = 2$ means $b = a^2$.

why: $\log_a b = c \Leftrightarrow a^c = b$.

2 $\log_b c = 3$ means $c = b^3$.

why: Same conversion with the next pair.

3 Substitute: $c = (a^2)^3 = a^6$.

why: Powers of powers multiply.

4 Therefore $\log_a c = 6$.

why: Read the exponent straight off. (In general $\log_a c = \log_a b \times \log_b c = 2 \times 3$.)

Where it goes wrong

- Adding the values to get 5.
- Reading $(a^2)^3$ as a^5 .

Takeaway Logarithms chain by multiplication: $\log_a c = \log_a b \times \log_b c$.

Question 3 LOG-06 · Exponentials and logarithms · spec MM5.1

The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

Answer: A

$$y = 2^{x-3} = 2^{-3} \cdot 2^x = \frac{1}{8} \cdot 2^x. \text{ (A translation of an exponential is also a stretch.)}$$

Full method - how and why

A translation of an exponential graph is also a stretch. The algebra shows why.

- 1 Translating 3 right replaces x by $x - 3$: $y = 2^{x-3}$.

why: A shift to the right by a substitutes $x - a$.

- 2 Split the index: $2^{x-3} = 2^x \times 2^{-3} = \frac{1}{8} \cdot 2^x$.

why: $a^m \cdot a^n = a^{m+n}$ - the translation has become multiplication by a constant.

- 3 So the new graph is the old one stretched vertically by factor $\frac{1}{8}$.

why: This is special to exponentials; horizontal translation and vertical stretch are the same thing.

Where it goes wrong

- Writing $2^{x-3} = 2^x - 3$ and choosing that option.
- Using $2^{+3} = 8$ as the factor, which corresponds to a shift to the **left**.

Takeaway $a^{x-k} = a^{-k}a^x$: for exponentials, translating horizontally is the same as stretching vertically.

Question 4 LOG-08 · Exponentials and logarithms · spec MM5.3

Given that $5^x = 2$ and $2^y = 125$, find the value of xy .

Answer: B

$$5^{xy} = (5^x)^y = 2^y = 125 = 5^3, \text{ so } xy = 3.$$

Full method - how and why

Two exponential facts that chain together. Raising the first to the power y links them.

- 1 Raise the first equation to the power y : $(5^x)^y = 2^y$, i.e. $5^{xy} = 2^y$.

why: $(a^x)^y = a^{xy}$ - this is what connects the two statements.

- 2 Substitute the second fact: $2^y = 125$.

why: Given directly.

- 3 Write 125 as a power of 5: $5^{xy} = 5^3$.

why: Both sides are now powers of 5.

- 4 Equate the indices: $xy = 3$.

why: Valid because 5^u is one-to-one.

Where it goes wrong

- Trying to compute x and y separately as logarithms and getting stuck without a calculator.
- Multiplying the two equations to get $10^{x+y} = 250$, which leads nowhere.

Takeaway To find a **product** of exponents, raise one equation to the other's exponent and chain the results.

Question 5 LOG-14 · Exponentials and logarithms · spec MM5.3

The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is

Answer: A

$2^9 = 512 < 1000 < 1024 = 2^{10}$. Need $n \leq 9$ and $n + 3 \geq 10$, so $n = 7, 8, 9$: three integers.

Full method - how and why

A double inequality in powers of 2. Locate 1000 between consecutive powers, then translate into conditions on n .

1 Find where 1000 sits: $2^9 = 512$ and $2^{10} = 1024$, so $2^9 < 1000 < 2^{10}$.

why: Powers of 2 up to 1024 are worth knowing by heart.

2 The left condition $2^n < 1000$ requires $n \leq 9$.

why: $2^{10} = 1024 > 1000$, so $n = 10$ fails.

3 The right condition $1000 < 2^{n+3}$ requires $n + 3 \geq 10$, i.e. $n \geq 7$.

why: $2^9 = 512 < 1000$, so $n + 3 = 9$ (i.e. $n = 6$) fails.

4 Combine: $7 \leq n \leq 9$, giving $n = 7, 8, 9$ - three integers.

why: Count the integers in the range, inclusive.

Where it goes wrong

- Counting $9 - 7 = 2$ instead of 3.
- Using strict inequalities on n where the powers allow equality.

Takeaway Convert conditions on 2^n into conditions on n by comparing with the nearest powers, then count inclusively.

Question 6 LOG-17 · Exponentials and logarithms · spec MM5.3

The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .

Answer: E

With $y = 3^x > 0$: $y^2 - ky + 9 = 0$. Exactly one solution for x means exactly one positive root: a repeated root ($k^2 = 36$, product $9 > 0$ so the roots have the sign of k): $k = 6$ (repeated root $y = 3$, $x = 1$). $k = -6$ gives $y = -3$, no solutions. (Two distinct positive roots would give two solutions.)

Full method - how and why

A quadratic in 3^x with a parameter. 'Exactly one real solution' is a condition on the **positive** roots, not simply on the discriminant.

1 Substitute $y = 3^x$, noting $y > 0$: $y^2 - ky + 9 = 0$.

why: $9^x = (3^x)^2$. The restriction $y > 0$ is the heart of the question.

2 The product of the roots is $9 > 0$, so the roots have the same sign; their sum is k , so both are positive exactly when $k > 0$.

why: Sum and product of roots decide the signs without solving.

3 For exactly one value of x , we need exactly one positive root, which (given both roots share a sign) means a **repeated** root: discriminant $k^2 - 36 = 0$, so $k = \pm 6$.

why: Two distinct positive roots would give two solutions for x ; two negative roots would give none.

4 Check the signs: $k = 6$ gives the repeated root $y = 3 > 0$, so $x = 1$ - exactly one solution. $k = -6$ gives $y = -3 < 0$, so no solutions at all. Hence $k = 6$.

why: The final sign check is what rules out -6 , which is offered.

Where it goes wrong

- Answering ± 6 from the discriminant alone.
- Forgetting that y must be positive and counting $y = -3$ as a solution.

Takeaway With $y = a^x$, translate conditions on x into conditions on the **positive** roots in y , using sum and product to check signs.

Question 7 LOG-19 · Exponentials and logarithms · spec MM5.3

A quantity halves every 5 hours. Which expression gives the number of hours for it to fall to $\frac{1}{10}$ of its initial value?

Answer: D

$(\frac{1}{2})^{t/5} = \frac{1}{10}$. Taking logs to base 10: $-\frac{t}{5}\log_{10} 2 = -1$, so $t = \frac{5}{\log_{10} 2}$ (about 16.6 hours).

Full method - how and why

Exponential decay. Build the model, then take logs - and be careful which expression the answer takes.

1 Model: after t hours the quantity is multiplied by $(\frac{1}{2})^{t/5}$, because $t/5$ half-lives have passed.
why: Dividing t by the half-life counts how many halvings have occurred.

2 Set it equal to $\frac{1}{10}$: $(\frac{1}{2})^{t/5} = \frac{1}{10}$.
why: 'Falls to one tenth' is a statement about the multiplier.

3 Take $\log_{10}$: $\frac{t}{5}\log_{10} \frac{1}{2} = \log_{10} \frac{1}{10} = -1$, i.e. $-\frac{t}{5}\log_{10} 2 = -1$.
why: $\log \frac{1}{2} = -\log 2$ and $\log \frac{1}{10} = -1$; the two minus signs will cancel.

4 Solve: $t = \frac{5}{\log_{10} 2}$ (about 16.6 hours).
why: Multiply both sides by $-\frac{5}{\log_{10} 2}$.

Where it goes wrong

- Answering $5\log_{10} 2$ - the reciprocal, which would be under a second.
- Sanity check: the quantity halves in 5 h and $\frac{1}{10} < \frac{1}{8}$, so t must exceed 15 h; $5\log_{10} 2 \approx 1.5$ is obviously wrong.

Takeaway Set up the multiplier, take logs, and sanity-check the size of the answer against a couple of half-lives.

Question 8 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim 'for all positive x and y ,

$\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$ '?

Answer: E

$x = y = 1$: $\log 2 > 0 = \log 1 + \log 1$, so the inequality fails. For the others $x + y \leq xy$ and the claim holds (log is increasing).

Full method - how and why

To break an inequality claim, look for the case where the two sides are closest - here small values, where logs go to zero or negative.

1 Test $x = y = 1$: left side $\log_{10} 2 \approx 0.30$; right side $\log_{10} 1 + \log_{10} 1 = 0$.
why: $\log 1 = 0$ for any base, so the right side collapses.

2 $0.30 > 0$, so the claimed inequality $\leq$ fails. **Counterexample.**
why: Hypothesis satisfied (x, y positive), conclusion false.

3 Why the others work: $\log x + \log y = \log(xy)$, and log is increasing, so the claim is just $x + y \leq xy$. For $x = y = 2$: $4 \leq 4$ ✓; $x = y = 4$: $8 \leq 16$ ✓; $x = y = 5$: $10 \leq 25$ ✓; $x = 10, y = 100$: $110 \leq 1000$ ✓.
why: Converting the log inequality into an algebraic one makes all five cases quick to check.

Where it goes wrong

- Trying large values, where xy dwarfs $x + y$ and the claim comfortably holds.
- Forgetting $\log_{10} 1 = 0$ and miscomputing the right side.
- Thinking $\log(x + y) = \log x + \log y$ - that is the error the whole question is built on.

Takeaway $\log x + \log y = \log(xy)$, never $\log(x + y)$. Test inequality claims near the boundary, where the sides are closest.

Question 9 X2-22 · Exponentials and logarithms · spec MM5.3

Consider the statement: **for all real x , $2^x > x^2$** . Which of the following is a counterexample?

Answer: B

$2^3 = 8 < 9 = 3^2$, so $x = 3$ breaks the claim. The other options satisfy it: $2 > 1$, $32 > 25$, $1 > 0$, $1024 > 100$.

Full method - how and why

Exponential versus quadratic growth: the exponential eventually wins, but there is a window where it loses. Test values inside that window.

- 1 Check the candidates: $x = 0$ gives $1 > 0$ ✓; $x = 1$ gives $2 > 1$ ✓; $x = 5$ gives $32 > 25$ ✓; $x = 10$ gives $1024 > 100$ ✓.

why: Evaluate both sides for each - no cleverness needed.

- 2 $x = 3$: $2^3 = 8$ and $3^2 = 9$. Since $8 < 9$, the claim fails. **Counterexample.**

why: Hypothesis (x real) satisfied, conclusion false.

- 3 Why here? 2^x and x^2 cross at $x = 2$ (both 4) and again near $x = 4$ (both 16), so the quadratic is ahead between them.

why: Knowing where the curves cross tells you exactly where to hunt.

Where it goes wrong

- Testing only large x , where the exponential dominates and the claim holds.
- Overlooking the window: the two sides are equal at $x = 2$ and $x = 4$, so the quadratic leads strictly between them - that is where a counterexample must lie.
- Assuming exponentials always beat polynomials - true only eventually.

Takeaway 2^x and x^2 meet at $x = 2$ and $x = 4$; between them the quadratic is larger. 'Eventually bigger' is not 'always bigger'.